

Ultra-Lightweight Network for Medical Image Segmentation Inspired by Bio-Visual Interaction

Zhefei Cai^{ID}, Yingle Fan^{ID}, Minwei Zhu, and Tao Fang^{ID}

Abstract—Computer-aided medical image segmentation helps to assist physicians in locating lesion area for the subsequent diagnosis and treatment. Due to the irregular shape of the target and the uneven sample size between the target and the background area, automatic segmentation of medical images is a challenging task. Many CNN-Based, Transformer-Based models deepen the number of network layers or introduce complex modules in order to improve the segmentation accuracy. Limited by the computational resources, these types of large models are not suitable for the actual clinical environment. Inspired by the rapidity, accuracy, and low consumption characteristics of bio-visual processing, the Ultra-Lightweight Network Inspired by Bio-Visual Interaction (BVI-Net) is constructed in this paper. The Global Pathway is constructed by simulating the dorsal stream, in order to extract global features rapidly, and the Local Pathway is constructed by simulating the ventral stream, in order to process local features finely. At the same time, the skip connection module integrating Graph Convolutional Network (GCN) attention mechanism is constructed to simulate the synchronous integration ability of the visual pathway for multi-level features. The International Skin Imaging Collaboration (ISIC) dataset, the Liver Tumor Segmentation (LiTS) dataset, and the Brain Tumor Segmentation Challenge (BraTS) dataset are used for experiments. The BVI-Net proposed in this paper requires only 0.026M parameters to achieve the excellent performance in three representative medical image segmentation datasets, which has certain advantages over state-of-the-art (SOTA) methods. The biological vision mechanism and the artificial intelligence algorithm are integrated in this paper, which provides new ideas for the construction of biological vision-guided deep learning models and promotes the development of biomimetic computational vision.

Index Terms—Biological visual pathway, dual encoder architecture, GCN attention mechanism, state space models, medical image segmentation.

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I. INTRODUCTION

MEDICAL imaging is an indispensable and important part of the medical field. Through medical imaging techniques such as Computed Tomography (CT) [1], Ultrasound (US) [2], and Magnetic Resonance Imaging (MRI) [3], organ or lesion areas are extracted from the images, which can help clinicians to observe the diseased parts clearly and quickly. With the development of computer technology, computers are gradually able to replace manual medical image segmentation tasks. Traditional machine learning methods, including the methods of threshold [4], region [5], clustering [6], etc., use features such as color and shape to complete the segmentation task. But they are only suitable for simple tasks where the target background difference is obvious, and the target boundaries are clear. Also, there are certain deficiencies in detection accuracy. The deep learning methods use self-supervised methods [7] like Convolutional Neural Networks (CNN) convolution module [8] and the Transformer attention module [9] to achieve the effect of pixel set segmentation through training iteratively. However, when facing the tasks of medical image segmentation with a lack of data and low data quality, data-driven deep learning methods face the challenges of detection accuracy versus sample quantity/quality, as well as detection accuracy versus computational consumption [10].

At present, there are three types of challenges in medical image segmentation tasks: (1) Difficulty in accurate segmentation due to blurred lesion margins. (2) Difficulty in global-local balance due to large differences in the extent of lesions. (3) Overfitting of deep complex networks due to lack of data. In order to solve the problem of edge blurring, the orientation selective characteristic of the receptive field (RF) in primary visual cortex (V1) is simulated in this paper. Gabor function is used to guide the initialization of convolutional kernels, in order to enhance the orientation information, which can help to complete the accurate localization of the lesion margins. Aiming at the difficult problems of global-local balance, the Global Pathway for rapid extraction of global features and the Local Pathway for fine extraction of local features are inspired by the structure of dorsal stream and ventral stream of biological vision. The dual pathways are processed in parallel and interacted between layers to ensure the complete extraction of global and local features. In order to solve the problem of overfitting, the topological mapping structure of the visual cortex is referred, the multi-level interconnection

structure of visual cortex is simulated, and the neural network model in accordance with to the visual information flow is constructed, which reduces the number of network layers as much as possible, trimming the redundant modules, and ensuring lightweight. Based on the above modules, the Ultra-Lightweight Network Inspired by Bio-Visual Interaction (BVI-Net) is proposed in this paper, and the main contributions are summarized as follows:

1. Simulating the orientation selective characteristic of the RFs in V1, Gabor function that fuses the “Cardinal Bias” and “Radial Bias” is used to guide the initialization of the convolutional kernel, making the convolutional kernels more sensitive to the orientation information.
2. Inspired by the parallel mechanism of bio-visual dual pathways and the multi-level interconnection, the dual encoder structure is constructed to extract the global and local features. The global information obtained by the Global Pathway is used to guide the Local Pathway to locate the lesion area, which helps the network converge quickly.
3. The topological mapping structure of bio-visual cortex is simulated, which enables BVI-Net to balance the detection accuracy and the model complexity. Benefit from the weight sharing setting of the Gabor convolution kernels in the Local Pathway, BVI-Net can achieve SOTA results with only 0.026M parameters and 0.025 GFLOPs, which provides more possibilities for the construction of biological vision-guided deep learning models.

II. RELATED WORKS

A. CNN-Based Segmentation Methods

Inspired by the excellent performance of CNN in image classification tasks, Ronneberger et al. [11] used convolution modules to construct an encoder-decoder structure named U-Net, which completed the image segmentation task by continuously down-sampling and up-sampling. Because of its simple structure and fast training, U-Net model is widely used in medical image segmentation tasks. Due to the existence of the skip connection, U-Net can supplement the target features appropriately in the up-sampling process, but the medical image segmentation tasks have pixel-level accuracy requirements, the down-sampling process of U-Net is easy to cause the loss of shallow features while extracting the deep features. In order to improve the ability to complement shallow features during up-sampling of U-Net, Zhou et al. [12] proposed the UNet++ model based on nested sampling, which improved the connectivity of model through dense skip connections. Huang et al. [13] proposed the UNet3+ model with full-scale skip connections, which fused low-level detail features with high-level semantic information to complement the missing features in the down-sampling process. Ibtehaz et al. [14] added the deep convolutional modules in skip connections to construct the MultiResUNet, in attempt to convert the local semantic information into higher-level global features.

The addition of skip connection module helps the network to retain more shallow features, but there are still

shortcomings in detection accuracy when facing the task of medical image segmentation with complex shapes. In order to improve the detection accuracy, Ramadan et al. [15] proposed the DGCU-Net with dual encoder structure to extract the color features and boundary features respectively to enhance the detection effect of lesion edges. Jha et al. [16] superimposed two U-Net networks to propose the DoubleUNet, with two sampling processes to extract more semantic information. Fakhry et al. [17] proposed the Residual Deconvolutional Network (RDN) based on U-Net, in which residual paths were added to the deconvolution network to obtain full-resolution features and contextual information, respectively. Li et al. [18] focus on the context information extraction module, designing a Differential Regional Attention mechanism to focus on extracting high-level semantics, so as to optimize the boundary regions. They want to achieve the lightweight effects, but the model still had a lot of parameters, and it could not reach the ultra-lightweight level. Ma et al. [19] broke the traditional U-Net symmetry model and proposed an Ultra-lightweight Fully Asymmetric Convolutional Network (ULFAC-Net), which used the parallel asymmetric convolution module with the dual attention structure to extract features, as well as enhancing the multi-level feature extraction to optimize the edge details. Considering that the same type of infected region or lesions often has a diversity of scales. Zhou et al. [20] presented a novel Uncertainty-aware Hierarchical Aggregation Network, proposing an Uncertainty-induced Cross-level Fusion module to fully fuse the features from the adjacent levels.

However, U-Net series models have the problems of global feature fusion, which are common in the convolutional networks. They lack the comprehensive consideration of global information, converge slower in the processing of features, and are easy to fall into the local optimal solution.

B. Transformer/Mamba-Based Segmentation Methods

Transformer-based models with global self-attention mechanism make the network pay more attention to the important regions of interest and have better processing ability for the overall information by simulating the human visual attention mechanism. The proposed of Vision Transformer [21] extended the application of Transformer from natural language processing to computer vision. To address the limitations of CNN in capturing global contextual information, Wu et al. [22] proposed the Feature-adaptive Transformer Network (FAT-Net), which can obtain the global contextual information effectively. Wang et al. proposed the Boundary-aware Transformer [23] and XBound-Former [24] successively, which utilized the global self-attention mechanism of Transformer to construct the boundary learner, focusing on the extraction of edge features and fusion with global information. Wang et al. [25] proposed the UCTransNet, which used Transformer to cross-fertilize the multi-scale channel features and utilized the feature maps at each resolution to solve the problems of semantic gap caused by the long connections. Liu et al. [26] proposed a Trans former network called CrackFormer, using content-based self-attention blocks

to exploit the global context for road surface crack detection. Chen et al. [27] proposed the LECSFormer, employing the window-based transformer blocks to model the long-range dependencies, and the dense connections are used to fuse the multi-scale information.

Transformer-based models show excellent performance in global feature extraction, but they are computationally complex when facing intensive prediction tasks such as medical image segmentation. Moreover, the detection accuracy of Transformer series models is extremely correlated with the amount of data, large amounts of high-quality data need to be used for training. Gu et al. [28] proposed Mamba based on State Space Models (SSM), which not only established long-distance dependencies, but also exhibited linear complexity in input size, saving a lot of computational cost compared with Transformer. Based on this, Ruan et al. [29] proposed VM-UNet for medical image segmentation tasks, introducing Visual State Space (VSS) blocks to capture the contextual information. Wu et al. [30] proposed the Parallel Vision Mamba (PVM) from the perspective of lightweight, which can complete the medical image segmentation task with very low computational complexity through the process of channel slicing. However, in the face of the tasks of medical image segmentation with small targets, the Mamba-based models cannot balance the extraction of global-local features, and it is easy to ignore the local detail features, causing the training process cannot converge.

C. Hybrid Segmentation Methods

In view of the global information extraction problem of CNN and the loss of detailed features of Transformer, researchers have tried to combine CNN and Transformer. Chen et al. [31] merged both Transformers and U-Net to propose the TransUNet, using Transformer to encode the tokenized image patches from CNN feature map in the encoder for extracting the global contexts, and upsampling the encoded features combined with the high-resolution CNN feature maps in the decoder to enable precise localization. Cao et al. [32] proposed the SwinUNet, which is an Unet-like pure Transformer for medical image segmentation tasks. U-shaped Encoder-Decoder architecture is responsible for extracting local features, and the hierarchical Swin Transformer with shifted windows are used to extract the context features. Regarding the situation of hyperspectral medical images, Yun et al. [33] proposed a spectral transformer for hyperspectral pathology image segmentation, which can capture the global context for intra-scale spectral features, and then extract the coarse and fine-grained discriminative spectral information from inter-scale features. Aiming at the small sample image segmentation tasks, Qi et al. [34] brought together CNNs and Transformers in a serial or interleaved architecture in order to incorporate long-range dependencies into the features extracted using CNNs. Although it can compensate for some of lost detail information in the Transformer, it also introduced too many parameters, increasing the training burden. Qiu et al. [35] explored the combination of Transformer and CNN to learn both global and local representations,

and proposed the transformer-based Asymmetric Bilateral U-Net (ABiU-Net), helping to detect the salient object effectively. However, the use of Transformer greatly increases the computational complexity of the model, which is not suitable for the task of small-sample medical image segmentation. Liao et al. [36] and Xing et al. [37] using Mamba instead of Transformer, proposed LightM-UNet and SegMamba, integrating Mamba and UNet to extract the deep semantic features and model the long-range spatial dependencies. Although these methods only need about 1m of parameters to complete the task of medical image segmentation, it still cannot achieve the ultra-lightweight, and the segmentation accuracy can be improved.

III. METHODS

Biological vision system has the multi-level interconnection structure, which can perceive and obtain different features in the input information through different pathways, process them in parallel and transmit them into higher visual cortex to complete the summary fusion and cognitive understanding [38]. Thanks to the dual pathway structure of dorsal stream and ventral stream, primates are able to balance the rapidity and accuracy of visual information acquisition [39]. Dorsal stream has visually salient attention characteristics, is sensitive to global features, and can quickly obtain the overall features, which is mainly processing the spatial position information [40], [41]. Ventral stream is sensitive to detail information and is mainly responsible for object recognition, which is not as fast as dorsal stream, but it is good at extracting local detail features [42].

In this paper, the structure of dorsal-ventral stream is simulated. Global Pathway is constructed to simulate the Dorsal stream, using State Space Models to extract the global features. At the same time, Local Pathway is constructed to simulate the Ventral stream, using Gabor convolution kernels to extract edge information, which helps to collect and refine the local features. Considering that the dorsal stream processes visual information faster than ventral stream, the information formed by it can be transmitted to ventral stream through the transverse channel, thus enhancing the response to visual stimuli at the corresponding location of the ventral stream [43]. The global features extracted by the Global Pathway are used to assist the Local Pathway to locate the lesion area through multiplication, so as to complete the fine extraction of edge features. Biological visual pathways are not completely independent, there are many lateral channels in them [44], contributing to the interactive integration of visual information by the bottom-up attention mechanism. Selective attention mechanism as an important attribute of primates to deal with complex visual field scenes, is capable of shifting attention to salient target areas according to the contextual information, and can capture the targets in different ways according to the differences of perceived features. Considering the disadvantages of CNN in dealing with global features, an attempt is made in this paper to convert the features extracted from Global Pathway and Local Pathway into graphs, incorporating the GCN Attention module into the skip connection to extract the comprehensive features in the graph through Graph

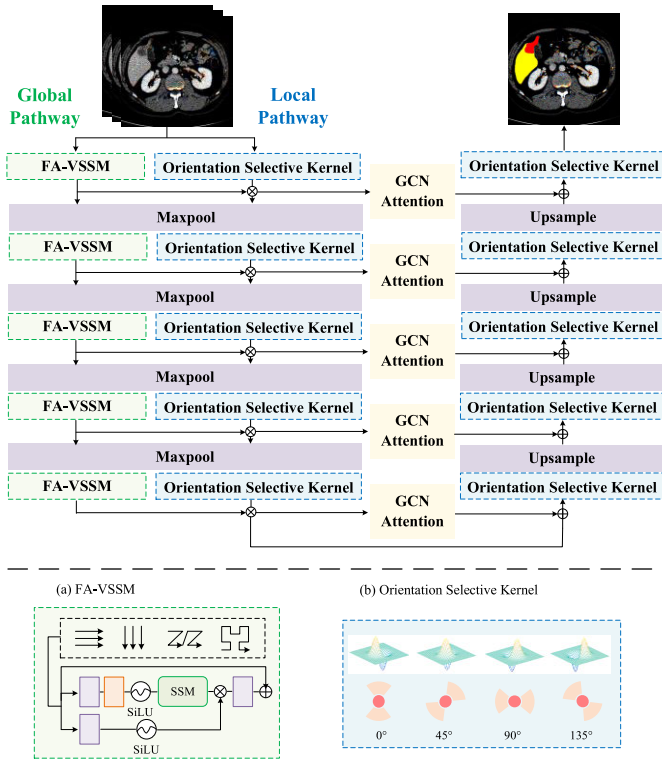


Fig. 1. Overall architecture of BVI-Net.

Convolution, so as to complete the effective fusion of feature information at all levels. During the upsampling phase, the features extract by the GCN Attention modules are added to the features of Local Pathway, so as to compensate for details that may be lost during upsampling. Combining the Local Pathway, Global Pathway and GCN skip connection module, BVI-Net is proposed in this paper, the overall Architecture is shown in Fig. 1.

A. Local Pathway Integrating Edge Information

The visual cortex of mammals presents the clear orientation preference, with most neurons responding only to a single orientation of their preference, and the function columns of different preferences are arranged orderly in space [45]. Each region composed of different functional columns is structurally differentiated and functionally divided [46], to extract and process the input visual information. According to Jones et al. [47], Gabor filter has the biological characteristics similar to RFs in V1, which is sensitive to the edge information and can effectively perceive edge features. Luan et al. [48] constructed the Gabor convolutional network to filter the input images through Gabor function, in order to extract the edge features more completely. However, the network does not fully consider the correspondence between Gabor function and RFs, using Gabor filtering simply is easy to cause the detail features losing. In order to better use Gabor function to simulate the RF characteristics, the Gabor convolutional kernel is constructed in this paper, which fuses Gabor function with convolutional kernel weights to guide the convolutional kernel initialization. The function of Gabor convolutional kernel is shown in (1).

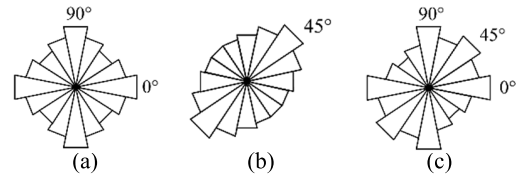


Fig. 2. Orientation distribution model of neurons in the visual cortex (a) The Model of Cardinal Bias (b) The Model of Radial Bias (c) The Model Combine Cardinal and Radial Bias.

When the stimulus reaches, the responses of neurons with different orientations to the stimulus $f(x, y)$ can be described by calculating the responses through Gabor filter in different orientations, as shown in (2).

$$\text{Gabor}(x, y; \theta; \sigma) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{\tilde{x}^2 + \gamma^2\tilde{y}^2}{2\sigma^2}\right) \cos\left(2\pi\frac{\tilde{x}}{\lambda} + \varphi\right) \quad (1)$$

$$h(x, y; \theta_i; \sigma) = \int \int f(x, y) \text{Gabor}(x, y; \theta_i; \sigma) dx dy \quad (2)$$

where, $(\tilde{x}, \tilde{y}) = (x \cos \theta + y \sin \theta, -x \sin \theta + y \cos \theta)$. θ represents the orientation of filter, which is the optimal direction of neuron response. By changing θ , it is possible to construct classical receptive field models with any orientations, in order to simplify the model, only four orientations $0^\circ, 45^\circ, 90^\circ, 135^\circ$ are used in the paper. γ represents the ellipticity, which is used to control the aspect ratio of filters. λ represents the wavelength.

Gabor convolution kernel is more in line with physiology, and the weight of the convolution kernel is adjusted adaptively with the training process, which not only contains the edge extraction ability of the Gabor function, but also avoids the problem of detail features losing caused by simple filtering. At the same time, the weights of different oriented convolution kernels in the same layer of the network are set to be shared, so as to reduce the parameters and computational consumption, and achieve ultra-lightweight requirement.

Fang et al. [49] used endogenous signal optical imaging method to study the orientation characteristics of neurons in the visual cortex of macaques, pointed out that the distribution of neurons preferring different orientations in the visual cortex are not uniform, existing the “Cardinal Bias” and the “Radial Bias”. Cardinal bias is manifested by a great number of neurons preferring horizontal and vertical orientations than the rest of orientations, and the radial bias, conversely, is manifested by a great number of neurons preferring diagonal orientations than horizontal and vertical orientation. Fig. 2 shows the model of cardinal bias, the model of radial bias, and the model combine cardinal and radial bias, respectively.

Biological visual system has the complex hierarchical structure, and the visual cortex can be divided into primary visual cortex (V1), secondary visual cortex (V2) and higher visual cortex (V4, MT and so on) according to the order of external input stimuli entering the visual cortex. The distribution of neurons in V1 and V2 can be expressed by the model combine cardinal and radial bias, with the modulation amplitude of 18%. The distribution of neurons in V4 can be expressed

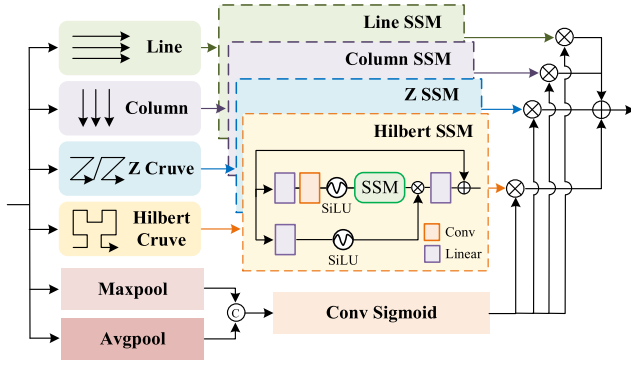


Fig. 3. The architecture of FA-VSSM.

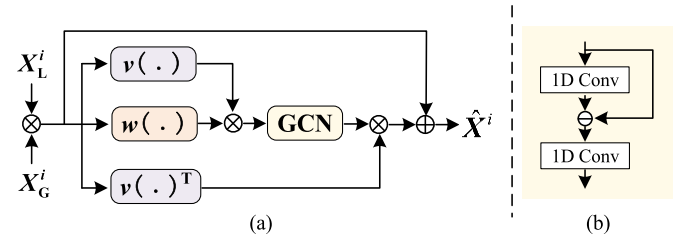
by the model of cardinal bias, with the modulation amplitude of 26%. This feature is considered with the ON/OFF characteristic of RF in this paper, constructing the edge modulation function as shown in (3)-(4), as shown at the bottom of the next page. It is fused with the convolutional kernel weights to construct the convolutional kernel with orientation selective characteristics. Setting the first 2 layers of Local Pathway represent the role of V1 and V2, using the model of combined bias, the last 3 layers of Local Pathway represent the role of V4, using the model of cardinal bias. where, $K_{V1/V2}$ represents the edge modulation function of RF in V1, V2. K_{V4} represents the edge modulation function of RF in V4. Gabor(.) represents the Gabor function. x, y represent the pixels in the input image. θ_1, θ_2 represent different orientations, $\theta_1 \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ\}$, $\theta_2 \in \{0^\circ, 90^\circ\}$. $\sigma_1, \sigma_2, \sigma_3, \sigma_4$ represent the size of the center RF in V1/V2, the size of the non-center RF in V1/V2, size of the center RF in V4, and the size of the non-center RF in V4, respectively. α, β represent the modulation amplitude. $\varepsilon_1, \varepsilon_2$ are random numbers, meeting $\varepsilon_1, \varepsilon_2 \in \{0,1\}$, $\varepsilon_1 + \varepsilon_2 = 1$.

B. Global Pathway Integrating State Space Models

Due to the presence of the dorsal stream, primates consciously focus their attention on the target area when dealing with complex visual scenes, while ignoring useless textured backgrounds [50]. Considering that CNN suffers from the dilemma of global features fusion, Transformer has the problem of high computational complexity. Compared with them, SSM can complete the global feature extraction task while maintaining linear complexity. In this paper, the fast extraction ability of dorsal stream for global features is simulated, the visual state space module integrating fast attention mechanism (FA-VSSM) is constructed, as shown in Fig. 3.

FA-VSSM consists of a fast attention channel and a state space channel. The fast attention channel consists of two parts: maximum pooling and average pooling, as shown in (5). It generates the adaptive attention coefficients in the channel dimension, sorts the importance of the input information, which assists the state space channel to filter the feature information.

$$FA_i = \text{Sigmoid} \left\{ \text{Conv2D} \left[\text{Concat} \left(\text{Avgpool} (I_i), \text{Maxpool} (I_i) \right) \right] \right\} \quad (5)$$

Fig. 4. The architecture of GCN attention module (a) The architecture of i -th block (b) The specific structure of graph convolution.

where, I_i represents the input of the i -th network. FA_i represents the adaptive coefficient of the i -th network.

The state space channel consists of a scan modules and SSMs [30]. Scan module is responsible for converting the input image into the one-dimensional sequence. Considering the connection between pixels, four conversion methods: line-by-line, column-by-column, Z-curve, and Hilbert curve are used in this paper, to increase the interaction information between different pixels. Finally, the adaptive attention coefficient is used to guide the SSMs to process the obtained feature sequences, and the scanning and merging operations are used to restore them to the initial size.

C. Skip Connection Integrating GCN Attention Mechanism

In order to obtain the global feature information, Chen et al. [51] proposed the Graph-based Global Reasoning module by using the global inference feature of graph convolution, but the image-to-graph process is too cumbersome, and needs pre-trained, which cannot meet the requirements of rapidity. In this paper, GCN Attention module is constructed by using the excellent extraction ability of graph convolution to extract the global features, as shown in Fig. 4. Multi-stage and multi-scale global features are extracted by graph convolution integrating the attention mechanism, and the encoder extracted features are screened during the process of skip connection, so as to highlight the key features and suppress the redundant noise.

The feature map shown in (6) is constructed in this paper. The feature information obtained by each layer of the dual pathway is projected from the coordinate space to the graph space, transforming the remote inference relationship in the feature map into the node inference relationship in the graph, so as to connect the non-adjacent regions in the coordinate space and integrate the pixels with similar features. In the mapping process $X \in \mathbb{R}^{H \times W \times C} \rightarrow G \in \mathbb{R}^{N \times C}$, the number of graph nodes N is set as a learnable parameter, which can be adjusted adaptively according to different datasets, avoiding the error caused by artificial setting of parameters.

$$G = v(X; W_v) \otimes \omega(X; W_\omega) \quad (6)$$

where, X represents the feature map fused by the Global Pathway features and the Local Pathway features. $v(\cdot), \omega(\cdot)$ represent the Graph mapping convolutional operations. W_v, W_ω represent the parameters of the convolution operation. $\otimes$ represents the matrix multiplication.

Feature mapping transforms the complex feature information represented by pixels into graph information represented

TABLE I
TOTAL NUMBER AND DIVISION OF DATASETS

Dataset	Training Set	Validation Set	Testing Set
ISIC2016	900	379	379
ISIC2017	2000	150	600
ISIC2018	2594	100	1000
ISIC2018-a	648	25	250
LiTS17	93/17046	13/775	26/1342
BraTS19	53/3584	8/472	15/912

Note: A/B style: the number of 3D images/ the number of 2D slices

by nodes and edges. Each node corresponds to a certain region feature in the original image, thus capturing the relationship between two regions in image is simplified to capturing the relationship between two nodes. Then, graph convolution is used to extract the features from the nodes and channels, as shown in (7). Finally, the graph is remapped to the original spatial domain, and used in the skip connection module to add the supplemental detail features, as shown in (8).

$$\hat{G} = ((I - A_g) G) W_g \quad (7)$$

$$\hat{X} = X + v(X; W_v)^T \hat{G} \quad (8)$$

where, $I \in \mathbb{R}^{N \times N}$ represents the unit matrix. $A_g \in \mathbb{R}^{N \times N}$ represents the adjacency matrix, it is the same learnable convolutional parameter as W_g , which is used to learn the edge weight information representing the interrelationships of the nodes.

IV. EXPERIMENT

A. Datasets and Training Settings

In order to verify the validity and robustness of the model, the segmentation tasks of skin lesions, liver tumors and brain tumors are applied in this paper respectively. Table I details the amount of data used in training, validation, and testing in this paper.

ISIC2016 [52], ISIC2017 [53] and ISIC2018 [54] datasets are used for the segmentation task of skin lesions. Among them, the division of ISIC2016, ISIC2017 and ISIC2018 is from the official. Considering the time and the consumption of the experiments, we randomly select one fourth images from ISIC2018 dataset in the ablation studies, which named ISIC2018-a. All input images are scaled to 256×256 .

LiTS17 dataset [55] is used for liver tumor segmentation task, which is derived from the 2017 Liver Tumor Segmentation challenge. The LiTS17 dataset contains 131 3D CT images, which are randomly divided at the ratio of 7:1:2. Each 3D CT image is transformed into a 2D slice, of which 19,163 contains liver labels and 7,190 contains tumor labels. In order to balance the positive and

negative samples, 19,163 images containing liver labels are used for the segmentation tasks, reasonably discarding the interference images without labels. All input images are scaled to 448×448 .

BraTS 2019 dataset [56] is used for brain tumor segmentation task, which is derived from the Multimodal Brain Tumor Segmentation Challenge [57], [58]. It provides 3D MRI dataset with ground truth tumor segmentation labels annotated by physicians. The BraTS 2019 dataset contains 335 cases, including 259 HGG cases and 76 LGG cases. Each case with four 3D MRI modalities (T1, T1c, T2 and FLAIR) rigidly aligned. Considering the complexity of the BraTS19 dataset, 76 LGG patients are used in this experiment, and it was divided randomly in the ratio of 7:1:2. The annotations are combined into 3 nested sub-regions: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET). Each 3D CT image is transformed into a 2D slice, same as LiTS17 dataset, data without any labels is removed. All input images are scaled to 240×240 .

The experimental environment is PyTorch framework version 1.13.0, the test platform is a computer equipped with NVIDIA GeForce GTX 3090 GPU. The epoch for a single training session is 50, the batch-size settings for all datasets is 64. The early stop strategy is set up in the experiment, if the Dice of the validation set does not increase and the loss does not decrease within 5 epochs, it will trigger an early stop. AdamW is used as the optimizer to update the parameters, with an initial learning rate of 0.001. Combined loss function is used to complete the segmentation task, as shown in (9).

$$Loss = \lambda_1 L_{Bce} + \lambda_2 L_{Dice}$$

$$L_{Bce} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

$$L_{Dice} = 1 - \left(2 \times \sum_{i=1}^N y_i p_i / \left(\sum_{i=1}^N y_i + \sum_{i=1}^N p_i \right) \right) \quad (9)$$

where, y represents the ground truth; p represents the prediction; λ_1, λ_2 represents the weight of Bce loss and Dice loss, setting $\lambda_1 = \lambda_2 = 1$.

Dice Score (Dice), mean Intersection over Union (mIoU), Accuracy (Acc), Specificity (Spe), Sensitivity (Sen) and Average symmetric surface distance (ASSD) are used to evaluate the segmentation results of all methods. To compare the parameters and computational complexity of each model, Parameters, Floating Point Operations (GFLOPs), and Frames Per Second (FPS) are calculated. It is important to note that the Params, GFLOPs, and FPS for each model are based on a single image with the input of 256×256 .

$$K_{V1/2} = \begin{cases} \text{Gabor}(x, y; \theta_1; \sigma_1) - [\alpha \cdot \text{Gabor}(x, y; \theta_2; \sigma_2) + \varepsilon_1 \cdot \text{Gabor}(x, y; 45^\circ; \sigma_2) + \varepsilon_2 \cdot \text{Gabor}(x, y; 135^\circ; \sigma_2)] \\ [\alpha \cdot \text{Gabor}(x, y; \theta_2; \sigma_2) + \varepsilon_1 \cdot \text{Gabor}(x, y; 45^\circ; \sigma_2) + \varepsilon_2 \cdot \text{Gabor}(x, y; 135^\circ; \sigma_2)] - \text{Gabor}(x, y; \theta_1; \sigma_1) \end{cases} \quad (3)$$

$$K_{V4} = \begin{cases} \text{Gabor}(x, y; \theta_1; \sigma_3) - \beta \cdot \text{Gabor}(x, y; \theta_2; \sigma_4) \\ \beta \cdot \text{Gabor}(x, y; \theta_2; \sigma_4) - \text{Gabor}(x, y; \theta_1; \sigma_3) \end{cases} \quad (4)$$

TABLE II
BASIC INFORMATION AND SEGMENTATION RESULTS OF ABLATION METHODS ON ISIC2018-A DATASET

Method	Params	GFLOPs	FPS	Dice	mIoU	Acc	Spe	Sen
Baseline	0.107	0.073	226.50	89.27	80.62	95.54	98.01	86.51
Local Pathway	0.019	0.015	312.41	89.63	81.31	95.67	97.94	87.34
Global Pathway	0.110	0.075	232.31	89.71	81.34	95.63	97.48	88.85
GCN Skip	0.112	0.082	215.71	89.58	81.13	95.63	97.84	87.55
Local + Global Pathway	<u>0.022</u>	<u>0.017</u>	<u>302.00</u>	90.24	82.22	95.92	<u>98.11</u>	87.91
Local Pathway + GCN Skip	0.024	0.023	301.75	90.09	81.96	95.75	<u>97.32</u>	89.99
Global Pathway + GCN Skip	0.114	0.084	184.90	<u>90.35</u>	<u>82.40</u>	<u>95.95</u>	97.99	88.64
BVI-Net	0.026	0.025	292.26	90.99	83.47	96.22	98.23	<u>88.88</u>

Note: **Bold** denotes the optimal performance; Underline denotes the suboptimal performance

TABLE III
BASIC INFORMATION AND RESULTS OF CONVERSION METHODS

Method	Params	GFLOPs	FPS	Dice	mIoU
Line	0.110	0.075	249.32	89.56	81.13
Column	0.110	0.075	249.32	89.54	81.09
Z-curve	0.110	0.075	<u>248.15</u>	89.55	81.10
Hilbert curve	0.110	0.075	<u>245.56</u>	<u>89.61</u>	<u>81.29</u>
Global Pathway	0.110	0.075	232.31	89.71	81.34

Note: **Bold** denotes the optimal performance; Underline denotes the suboptimal performance

TABLE IV
BASIC INFORMATION OF SOTA METHODS

Method	Year	Params	GFLOPs	FPS
U-Net	2016	13.395	31.119	51.42
MultiResUNet	2020	1.945	5.124	33.33
MALUNet	2022	0.178	0.083	105.89
UNeXt	2022	1.472	0.573	105.89
EGE-UNet	2023	0.053	0.072	90.01
VM-UNet	2024	44.274	7.562	29.32
MHorUNet	2024	4.962	0.629	123.07
UltraLight VM-UNet	2024	0.049	0.060	219.09
BVI-Net	--	0.026	0.025	292.26

B. Ablation Studies

ISIC2018-a dataset is used to evaluate the role of each module in BVI-Net. Local Pathway represents the edge feature extraction channel containing Gabor convolution, vanilla convolution is used to replace the Gabor convolution. Global Pathway represents the global feature extraction channel containing SSM, vanilla convolution is also used to replace the SSM module. GCN Skip represents the skip connection module containing GCN attention mechanism. The number of parameters and the segmentation performance of all ablation methods are shown in Table II.

It can be seen from Table II, Local Pathway with Gabor convolution has the lowest number of parameters, the lowest GFLOPs, and the highest FPS. Baseline, which increases the number of parameters and calculation time due to the use of vanilla convolution instead of Gabor convolution with shared parameters, is inferior to the Local Pathway in terms of segmentation results, thus demonstrating the importance of the local feature extraction module. Although Global Pathway and GCN Skip increase the number of parameters and calculation time, they improve the segmentation results. BVI-Net with edge/global feature extraction and GCN attention mechanism, is able to achieve the optimal segmentation results while the number of parameters and GFLOPs are lower than Baseline.

In order to verify the role of different conversion methods in the FA-VSSM, the Global Pathway is selected separately for the ablation studies. The computational cost and the segmentation performance are shown in TABLE III.

Line, Column, Z-curve and Hilbert curve represent the single conversion method, Global Pathway represents the combined of four conversion methods. It can be seen from Table III, the Params and the GFLOPs of the single conversion method and the Global Pathway are the same, because the number of conversion methods does not affect the

convolution kernels, so, there is no difference between the Params and the GFLOPs. However, it affects the FPS, the conversion methods of Line and Column with the highest FPS, indicating that these two methods consume the least amount of computation. As for the results of Dice and mIoU, Global Pathway performs best, with the improvement of 0.19% and 0.31%. Medical images are connected in the most areas, different conversion methods have little impact on global feature extraction, but for the independent small target lesion or the edge of the lesion, the fusion of different conversion methods is helpful to better locate the target and obtain more features.

C. Comparison With State-of-the-Art Methods

In order to evaluate the performance of BVI-Net, SOTA methods including U-Net [11], MultiResUNet [14], MALUNet [59], UNeXt [60], EGE-UNet [61], VM-UNet [29], MHorUNet [62], UltraLight VM-UNet [30] are compared in this paper. Except for U-Net and VM-UNet, the other methods are lightweight methods, and the basic information of each comparison method is shown in Table IV.

1) *Skin Lesion Segmentation Tasks:* For the segmentation of skin lesions, the results of each SOTA method on ISIC datasets are shown in Table V. It can be seen that BVI-Net proposed in this paper has the best performance, followed by VM-UNet and MHorUNet.

For Dice, BVI-Net shows 0.12%, 3.44%, and 0.47% improvement over the sub-optimal method. For mIoU, BVI-Net shows the improvement of 0.24%, 6.13%, 0.78%. And for ASSD, BVI-Net shows the improvement of 30.71%, 25.60%, 11.67%. Spe and Sen are the complementary indicators, BVI-Net has the better performance in Sen, but is slightly

TABLE V
RESULTS OF SOTA METHODS ON SKIN LESION SEGMENTATION TASKS

Dataset	Method	Dice	mIoU	Acc	Spe	Sen	ASSD
ISIC2016	U-Net	90.72	83.03	94.83	96.83	89.73	207.56
	MultiResUNet	87.56	77.87	92.79	93.85	90.09	547.38
	MALUNet	90.25	82.23	94.63	97.18	88.15	198.40
	UNeXt	91.33	84.04	95.09	96.43	91.69	<u>168.77</u>
	EGE-UNet	90.17	82.10	94.45	96.05	90.36	234.79
	VM-UNet	<u>91.87</u>	<u>84.95</u>	<u>95.39</u>	96.55	<u>92.42</u>	203.62
	MHorUNet	91.63	84.57	95.31	<u>96.89</u>	91.26	178.84
	UltraLight VM-UNet	90.72	83.14	94.79	96.52	90.37	226.11
ISIC2017	BVI-Net	91.98	85.15	95.46	96.65	92.43	116.94
	U-Net	80.48	67.34	<u>93.65</u>	97.48	75.42	431.30
	MultiResUNet	75.37	60.48	86.52	86.25	87.39	1776.38
	MALUNet	82.96	70.89	92.48	97.08	77.58	381.06
	UNeXt	83.51	71.69	92.61	96.72	79.30	409.57
	EGE-UNet	82.96	70.89	92.53	97.33	77.02	384.88
	VM-UNet	84.04	72.47	92.93	97.28	78.85	<u>356.50</u>
	MHorUNet	<u>84.54</u>	<u>73.21</u>	92.99	96.67	81.10	423.90
ISIC2018	UltraLight VM-UNet	83.55	71.75	92.76	<u>97.35</u>	77.89	369.08
	BVI-Net	87.45	77.70	94.19	96.79	<u>85.76</u>	265.24
	U-Net	86.47	76.16	94.15	96.09	87.06	450.95
	MultiResUNet	83.45	71.30	92.35	93.39	88.57	1185.02
	MALUNet	88.06	78.66	94.97	97.28	86.51	303.92
	UNeXt	88.70	79.69	95.27	<u>97.65</u>	86.54	288.40
	EGE-UNet	89.03	80.23	95.34	97.32	88.10	<u>234.79</u>
	VM-UNet	<u>90.00</u>	<u>81.81</u>	<u>95.72</u>	97.36	<u>89.74</u>	302.60
ISIC2018	MHorUNet	89.37	80.79	95.53	97.74	87.47	310.96
	UltraLight VM-UNet	89.16	80.45	95.38	97.21	88.66	248.04
	BVI-Net	90.43	82.45	95.86	97.15	91.14	207.63

Note: **Bold** denotes the optimal performance; Underline denotes the suboptimal performance

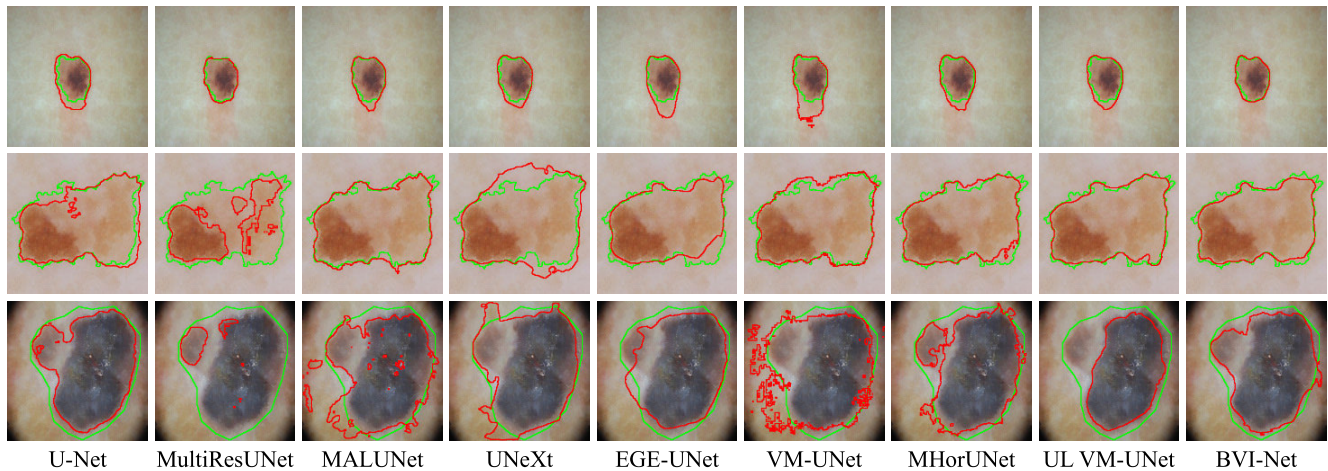


Fig. 5. Segmentation results of each SOTA methods (Green contours represent the ground truth and red contours represent the segmentation results).

insufficient in Spe, indicating that BVI-Net can segment the lesion area more completely, but it is prone to mistakenly segment the normal background into lesion areas. However, compared to the method of the optimal Spe, BVI-Net has 0.55%, 0.71% and 0.60% deficiencies in Spe, but has 4.86%, 13.71% and 4.20% improvement in Sen.

In order to show the segmentation performance of each SOTA method more intuitively, images for the three cases of small lesion area, similar lesion and background, blurred lesion edges are selected from the datasets for visualization and analysis, as shown in Fig. 5.

It can be seen from Fig. 5, for small lesions, most methods cannot solve the problem of positive and negative samples imbalance, and are easy to segment the lesion area too large, while BVI-Net and MultiResUNet can segment them more accurately. In view of the similarity between lesions and backgrounds, U-Net and MultiResUNet could not distinguish edges well, which make the segmentation results confusing. Also UNeXt, EGE-UNet, VM-UNet and MHorUNet have varying degrees of over- and under- segmentation. In contrast, BVI-Net, MALUNet, UltraLight VM-UNet have good performance. As for the situation of blurred lesion edges, MALUNet,

TABLE VI
RESULTS OF SOTA METHODS ON LIVER TUMOR SEGMENTATION TASKS

Method	Liver					Tumor				
	Dice	mIoU	Acc	Spe	Sen	Dice	mIoU	Acc	Spe	Sen
U-Net	98.09	96.25	99.65	99.78	98.37	70.50	55.45	99.74	99.92	64.00
MALUNet	97.46	95.05	99.54	99.75	97.41	76.85	62.41	<u>99.81</u>	99.96	67.42
UNeXt	97.92	95.93	99.62	99.77	<u>98.12</u>	<u>83.67</u>	<u>71.93</u>	99.85	<u>99.94</u>	<u>81.30</u>
EGE-UNet	92.91	86.75	98.69	99.36	92.17	31.84	18.94	99.78	99.93	25.90
VM-UNet	97.55	95.23	99.56	<u>99.78</u>	97.36	—	—	—	—	—
MHorUNet	95.33	91.44	99.13	99.51	95.62	54.29	37.26	99.64	99.90	45.10
UltraLight VM-UNet	95.67	91.70	99.22	99.66	94.84	—	—	—	—	—
BVI-Net	98.11	96.29	<u>99.63</u>	99.85	97.62	84.23	72.75	99.85	99.91	86.06

Note: **Bold** denotes the optimal performance; Underline denotes the suboptimal performance

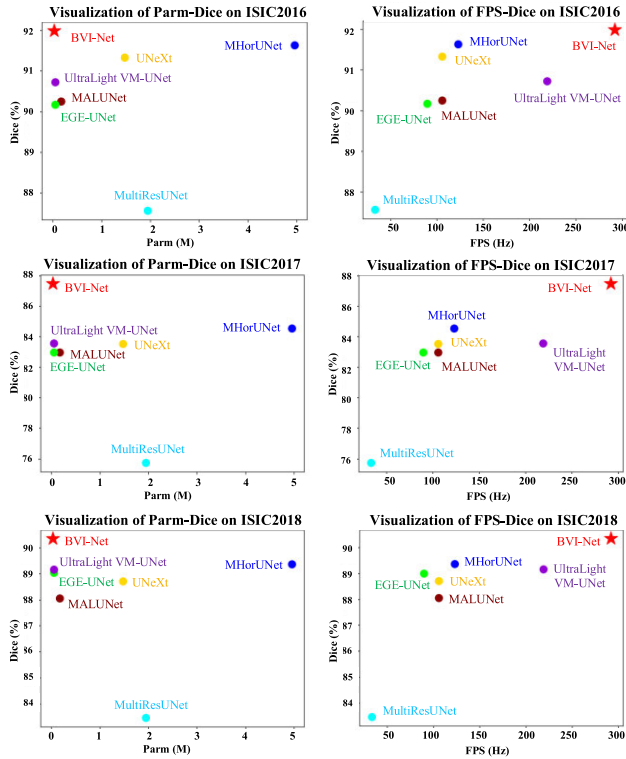


Fig. 6. Visualization analysis of each SOTA method.

VM-UNet, and MHorUNet having problems with segmentation confusion. MultiResUNet and UltraLight VM-UNet have more serious segmentation defects, MultiResUNet only segments the small lesion in the upper left corner, while UltraLight VM-UNet only segments the large lesion on the right side. Although BVI-Net and EGE-UNet performed better than other methods, there are still have some differences with labels, indicating that the problem of blurred lesion edges deserves further in-depth study.

At the same time, Params-Dice and FPS-Dice results of each SOTA method are visually analyzed on the ISIC datasets, as shown in Fig. 6. x-axis represents Params and FPS, respectively, the smaller the Params, the more lightweight the model, and the larger the PFS, the faster the processing speed. y-axis represents Dice, the higher the value, the better the segmentation result.

As can be seen from Fig. 6, in the Params-Dice visual analysis diagram, BVI-Net is located in the upper left corner, indicating that BVI-Net can achieve the optimal segmentation effect with the fewest parameters compared with other SOTA methods. In the FPS-Dice visual analysis diagram, BVI-Net is located in the upper right corner, indicating that BVI-Net can balance the segmentation accuracy and the processing speed.

2) *Liver Tumor Segmentation Tasks:* LiTS17 dataset is selected for liver tumor segmentation, which provides two labels: “Liver” and “Tumor”. The results of each SOTA method are shown in Table VI.

It can be seen that BVI-Net and UNeXt perform better for both the larger target “Liver” and the smaller target “Tumor”. Although U-Net has better segmentation results for “Liver”, it cannot accurately complete the segmentation requirements when facing “Tumor”. Due to the “Tumor” areas are too small, VM-UNet and UltraLight VM-UNet, which are good at processing global information, cannot converge during the training process because of lacking the ability to extract local features, so the segmentation task of “Tumor” cannot be completed.

In order to show the segmentation performance of each SOTA method more intuitively, images are randomly selected from dataset for visual analysis, as shown in Fig. 7.

It can be seen from Fig. 7 that each method can segment “Liver” well. Although UNeXt, EGE-UNet, VM-UNet, and MHorUNet mistakenly segment some non-liver regions into liver regions, the overall segmentation results are good. Compared with “Liver”, “Tumor” targets are smaller, there is an extreme imbalance between positive and negative samples, each method has shortcomings in the segmentation of “Tumor”. VM-UNet and UltraLight VM-UNet can only complete the tasks of “Liver” segmentation, while cannot deal with the tasks of “Tumor” segmentation. U-Net, EGE-UNet, and MHorUNet can complete the segmentation task when facing large “Tumor”, but cannot segment small “Tumor”, which explains why these methods perform poorly when segmenting “Tumor” in Table VI.

3) *Brain Tumor Segmentation Tasks:* BraTS19 dataset is selected for brain tumor segmentation, which provides three labels: “WT”, “TC”, and “ET”. The results of each SOTA method are shown in Table VII.

As can be seen from Table VII, BVI-Net can achieve the best segmentation results in three cases of “WT”, “TC”,

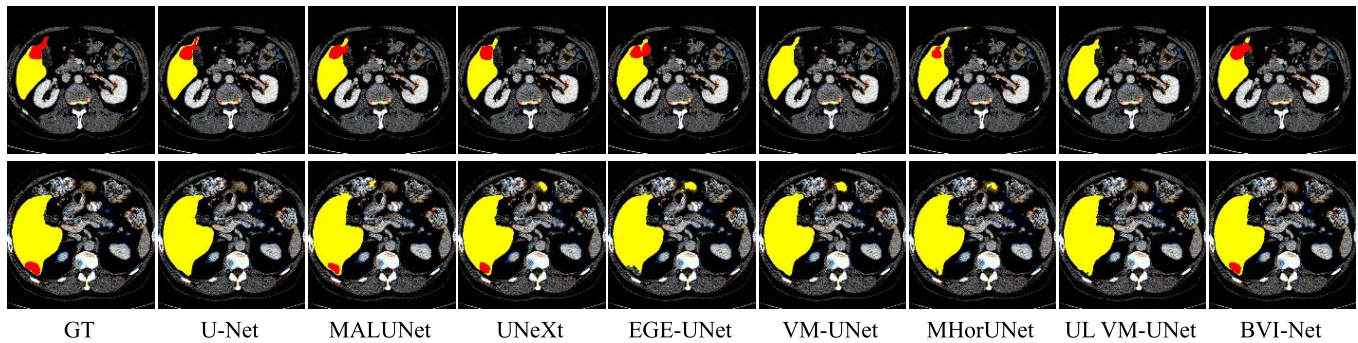


Fig. 7. Segmentation results of each SOTA methods (Yellow represents the liver and red represents the tumor).

TABLE VII
RESULTS OF SOTA METHODS ON BRAIN TUMOR SEGMENTATION TASKS

Method	WT		TC		ET		Average	
	Dice	mIoU	Dice	mIoU	Dice	mIoU	Dice	mIoU
U-Net	86.64	76.42	65.21	48.38	—	—	—	—
MALUNet	92.52	86.09	<u>88.94</u>	<u>80.08</u>	<u>67.79</u>	<u>51.27</u>	<u>83.08</u>	<u>72.48</u>
EGE-UNet	90.28	82.28	82.52	70.24	57.80	40.64	76.87	64.39
MHorUNet	<u>92.79</u>	<u>86.56</u>	88.13	78.79	58.63	41.47	79.85	68.94
UltraLight VM-UNet	88.69	79.68	66.66	50.00	45.68	29.60	67.01	53.09
BVI-Net	93.02	86.95	89.68	81.29	68.67	55.29	83.79	74.51

Note: **Bold** denotes the optimal performance; Underline denotes the suboptimal performance

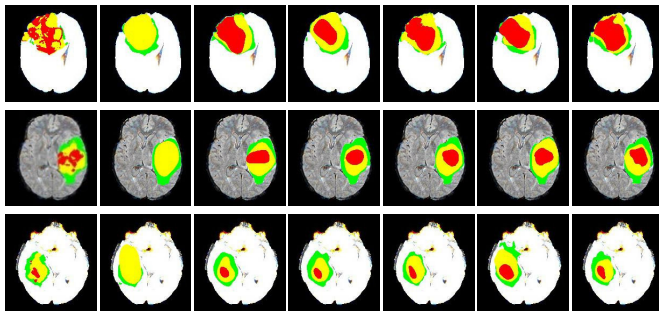


Fig. 8. Segmentation results of each SOTA methods. (GT, U-Net, MALUNet, EGE-UNet, MHorUNet, UL VM-UNet, BVI-Net respectively. Green represents WT, yellow represents TC and red represents the ET).

and “ET”. MHorUNet achieves sub-optimal results for “WT” category, and MALUNet achieves sub-optimal results for “TC” and “ET” categories. Due to the small size and the complex distribution of “ET” targets, U-Net cannot converge during the training process, so it cannot segment “ET” targets. In order to visualize the segmentation performance of each SOTA method when facing three types of targets “WT”, “TC”, and “ET”, images are randomly selected from dataset for visual analysis, as shown in Fig. 8.

It can be seen from Fig. 8 that U-Net can only complete the task of segmenting “WT” and “TC”, but not “ET”, which is the same as the results shown in Table VII. The targets area of “WT” and “TC” are larger, and the edge of the label are smoother, so each method can segment them well, but the targets region of “ET” are smaller, irregular and scattered, each method uses 2D slice after 3D slicing for training and testing, the segmentation results are more concentrated and the edges are smooth, which are different from GT.

V. CONCLUSION

Inspired by the multi-level interconnection structure of biological vision, dual-pathway encoder is constructed by simulating the rapid extraction ability of the dorsal stream and the fine processing ability of the ventral stream in this paper. Meanwhile, the features extracted by the global pathway are used as prior knowledge to guide the local pathway to locate the lesion area accurately, helping local pathway to better obtain the lesion edge features. In order to better integrate multi-level features, the skip connection module containing GCN attention mechanism is constructed in this paper, which fuses the multi-level features while extracting the global salient features. To verify the performance of BVI-Net proposed in this paper, experiments are conducted for three different application scenarios: skin lesions, liver tumors and brain tumors, respectively. Compared with the lightweight SOTA methods, BVI-Net not only minimizes the number of model parameters and maximizes the training efficiency, but also has the excellent performance on LiTS17, BraTS19 and ISIC series datasets. However, for the segmentation task of brain tumors, if small targets with discontinuous shapes are encountered, BVI-Net is easy to smooth their edges, which does not give the better segmentation results. It is necessary to focus on the targets discontinuity situation in the subsequent researches, and construct an appropriate network to optimize the segmentation effect. Based on the parameters, processing speed and segmentation performance, BVI-Net has certain advantages over the current lightweight SOTA methods. Moreover, BVI-Net integrates the biological vision mechanism into the construction of deep learning models, providing more possibilities for the construction of biological vision-guided deep learning models.

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